PowerGrid Utility Intelligence: Predicting Grid Failures Before They Happen

Final Technical Report — IIT Madras Capstone Programme

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1 Executive Summary

PowerGrid Utilities Corporation manages critical power infrastructure across five geographic regions. Aging assets, rising load demand, and increasingly severe weather events are driving up unplanned failure rates — each outage costing hundreds of thousands of dollars in direct losses plus regulatory penalties and reputational harm.

This project built a production-ready machine learning system that predicts which grid assets are likely to fail before failure occurs. Using a dataset of 50,500 asset records with 32 features, the final LightGBM model achieves:

- **Recall of 0.84** — meaning 84% of actual failures are flagged for preventive inspection
- **AUC-ROC of 0.9156** — strong overall discriminative ability
- **AUC-PR of 0.8982** — confirmed on the positive class (failures), the metric that matters

Key findings from the model:

- Equipment health score, composite risk score, and storm risk index are the three strongest predictors of failure
- 15,715 assets (31.4% of the portfolio) are classified as HIGH risk, representing $9.6 billion in revenue exposure
- Applying predictive maintenance to HIGH-risk assets could prevent $1.44B–$2.89B in outage losses annually at an inspection cost of approximately $23.6M — a 60× return

Business recommendation: Deploy the LightGBM scoring pipeline to produce weekly maintenance priority lists. Operations teams should schedule immediate inspections for the top 1,000 HIGH-risk assets (flagged with failure probability ≥ 0.90), transitioning from calendar-based to condition-based maintenance across all five regions within 12 months.

2 Business Understanding

2.1 Problem Context

PowerGrid Utilities Corporation operates transmission lines, transformers, substations, switchgear, and circuit breakers across five regions (Central, East, North, South, West) serving hundreds of thousands of customers. Unplanned equipment failures cause:

- **Direct losses:** outage restoration costs, emergency crew dispatch, equipment replacement
- **Regulatory penalties:** utility commissions impose fines for outage duration and frequency exceeding thresholds
- **Revenue impact:** estimated $250,000+ per major unplanned failure event
- **Customer harm:** service interruptions affect residential and industrial customers, with cascading supply-chain effects for the latter

Historically, maintenance has been scheduled on a calendar basis (e.g., inspect all transformers every 24 months). This approach is inefficient: it triggers unnecessary inspections of healthy assets while failing to prioritise degrading assets that fall between inspection windows.

2.2 Business and Analytical Objectives

Business objective: Reduce unplanned outages and financial exposure by proactively identifying high-risk grid assets for condition-based maintenance intervention.

Analytical objective: Build a binary classification model predicting `grid_failure_flag` (1 = failure, 0 = no failure) that maximises Recall — because the cost of a missed failure vastly exceeds the cost of an unnecessary inspection.

The model must also be explainable: operations engineers need to understand *why* a specific asset was flagged, not just *that* it was flagged. SHAP (SHapley Additive exPlanations) is used to provide feature-level attribution for every prediction.

2.3 Stakeholder Impact

Stakeholder	Pain point	How this project helps
Operations managers	Reactive maintenance, unpredictable crew schedules	Prioritised weekly inspection list
Finance / CFO	Unforecast outage costs, reserve underestimation	Quantified risk exposure per asset
Compliance / Legal	Regulatory penalty risk	Early identification of at-risk assets before breach
Customers	Service interruptions	Reduced outage frequency through preventive action

3 Data Understanding

3.1 Dataset Overview

Property	Value
Source	PowerGrid Utilities Corporation internal asset database
Records	50,500 (reduced to 50,000 after deduplication)
Features	32 raw features; 41 features after engineering
Target variable	`grid_failure_flag` (binary: 1 = failure, 0 = no failure)
Asset types	Transmission Lines, Transformers, Switchgear, Circuit Breakers, Substations
Regions	Central, East, North, South, West
Manufacturers	ABB, GE, Hitachi, Schneider, Siemens

Feature categories in the raw dataset:

- **Asset identity:** asset_id, asset_type, substation_region, manufacturer, installation_year, asset_age_years
- **Equipment condition:** equipment_health_score, oil_quality_score, transformer_temperature_c, vibration_level, insulation_resistance_kohm
- **Maintenance history:** maintenance_overdue_days, last_maintenance_days_ago, inspection_score, fault_history_count
- **Load and utilisation:** power_load_mw, load_utilisation_pct, peak_load_factor
- **Environmental exposure:** rainfall_mm, storm_risk_index, humidity_pct, temperature_c
- **Operational metrics:** avg_outage_duration_minutes, customers_served, estimated_revenue_loss, regulatory_penalty_cost

3.2 Class Distribution

The target variable is approximately balanced (43% failures, 57% non-failures), which avoids extreme imbalance while still warranting attention to Recall as the primary metric.

3.3 Failure Rates by Asset Type and Region

Failure rates are broadly similar across asset types (ranging from approximately 40% to 46%), with Transformers showing the highest absolute count of failures given they also have the highest asset count. Regionally, all five regions show failure rates within 2 percentage points of the overall mean — confirming no single geographic cluster drives the signal; the model must learn from equipment-level features rather than region alone.

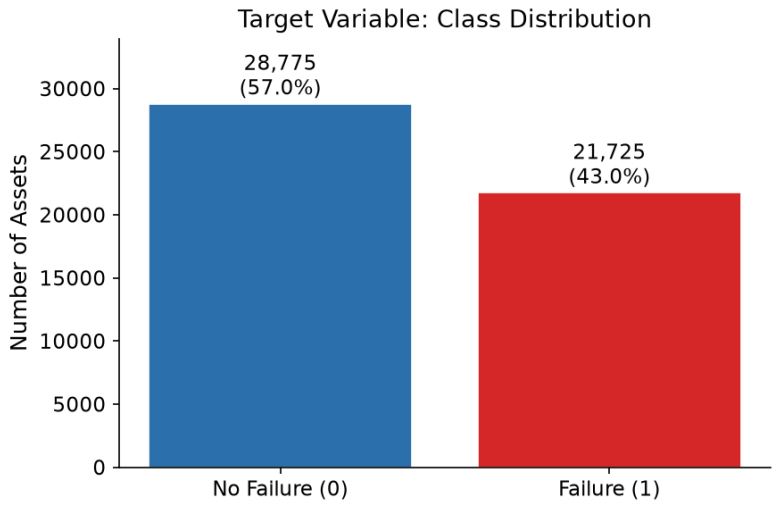


Figure 1: Class distribution of the target variable

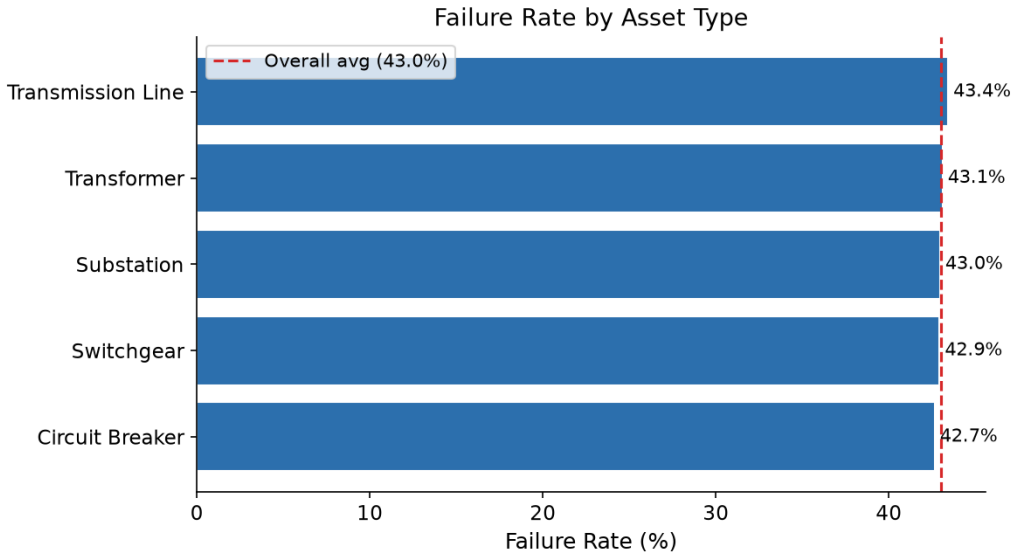


Figure 2: Failure rate by asset type

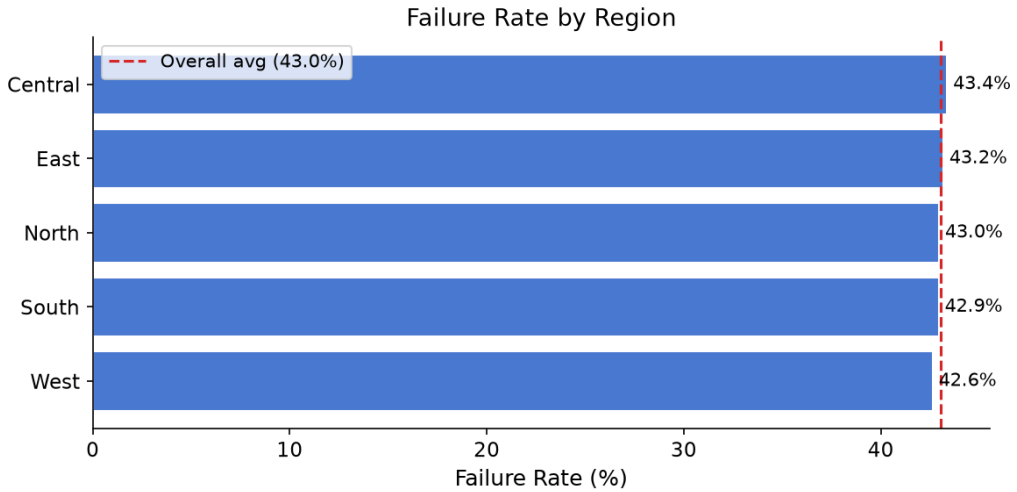


Figure 3: Failure rate by region

3.4 Data Quality Findings

Issue	Finding	Resolution
Duplicate rows	500 exact duplicates (~1% of dataset)	Removed; retained first occurrence
Missing values	`oil_quality_score`: 10.0% missing; `inspection_score`: 10.1% missing	Grouped median imputation by `asset_type`
Category casing	`asset_type` had three casing variants of "Transformer" (Transformer, transformer, TRANSFORMER)	Normalised with `.str.title()`
Outliers	`rainfall_mm`, `avg_outage_duration_minutes`, `power_load_mw` had ~1–1.4% extreme values	Winsorised at 1st/99th percentile (row deletion avoided to preserve failure events)
Collinearity	Equipment health, oil quality, transformer temperature expected to correlate	Addressed via VIF analysis (see Section 4)



Figure 4: Missing value summary

3.5 Feature Correlations with Target

The strongest positive correlates with failure include `maintenance_overdue_days`, `fault_history_count`, `storm_risk_index`, and `load_utilization_pct`. The strongest negative correlates (i.e., associated with healthy assets) include `equipment_health_score`, `insulation_resistance_kohm`, and `inspection_score`.

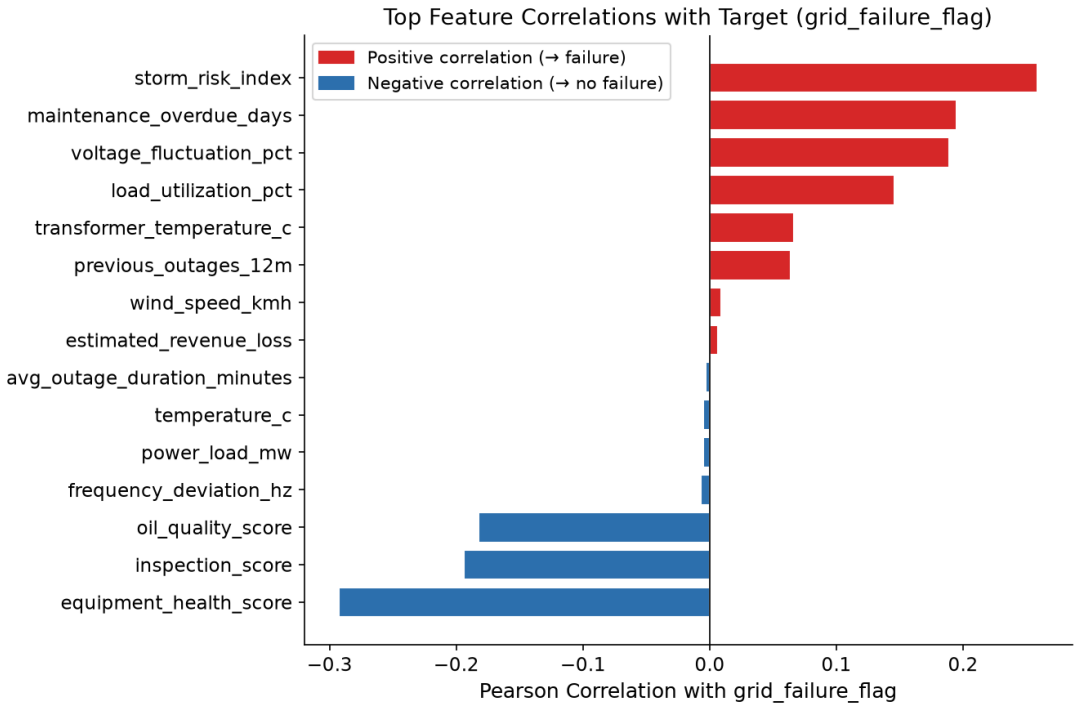


Figure 5: Top feature correlations with grid_failure_flag

3.6 Key Feature Distributions

Comparing failure and non-failure populations across the four most informative features reveals clear separation: assets that eventually fail tend to have lower equipment health scores, more maintenance overdue days, higher storm risk, and higher load utilisation. These patterns confirm that the raw features carry real predictive signal.

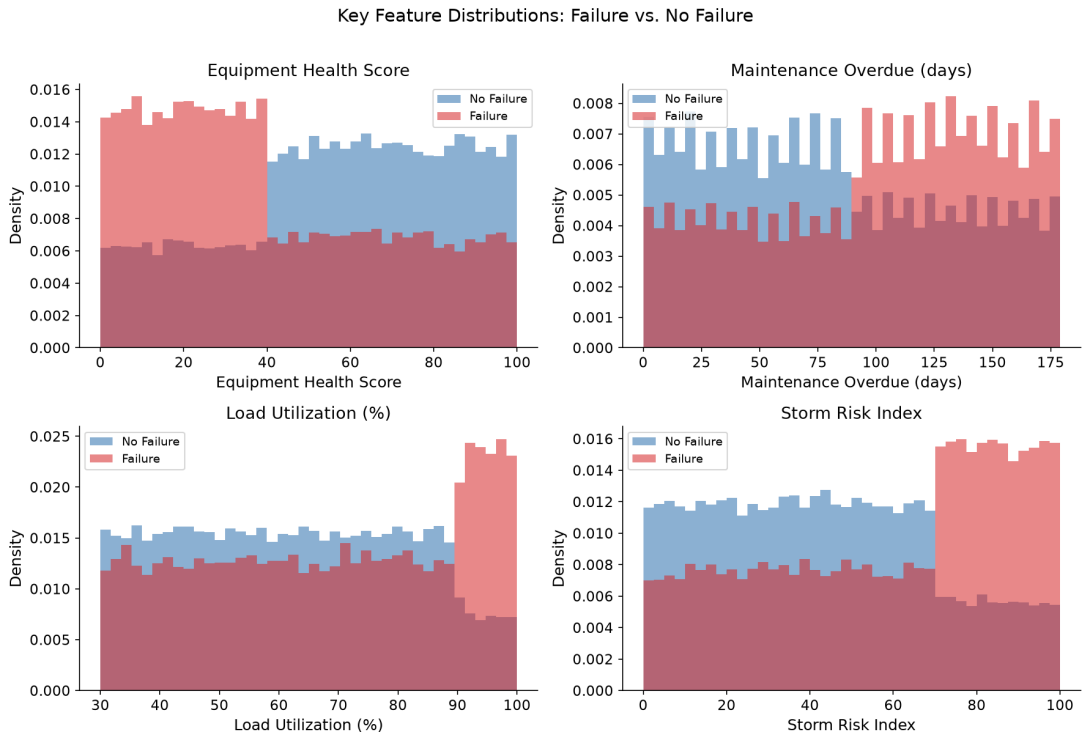


Figure 6: Key feature distributions by failure class

4 Data Preparation

4.1 Cleaning Steps

All cleaning was implemented in `notebooks/02_data_cleaning.ipynb` and in the standalone script `src/data_cleaning.py`.

Step 1 — Category normalisation: `asset_type` values were normalised with `.str.title()` which reduced the apparent 7 unique values (including casing variants) to the correct 5 asset types.

Step 2 — Deduplication: 500 exact duplicate rows were removed with `df.drop_duplicates()`, retaining the first occurrence. This reduced the dataset from 50,500 to 50,000 rows.

Step 3 — Missing value imputation: `oil_quality_score` (10.0% missing) and `inspection_score` (10.1% missing) were imputed using grouped median by `asset_type`. This approach is superior to global median imputation because asset health norms differ across types (a healthy Transformer has a different typical oil quality than a healthy Circuit Breaker).

Step 4 — Winsorisation: Three columns (`rainfall_mm`, `avg_outage_duration_minutes`, `power_load_mw`) with extreme outliers were Winsorised at the 1st and 99th percentiles. Row deletion was deliberately avoided to preserve all failure events; Winsorisation retains every row while capping extreme values.

Step 5 — Administrative column removal: `asset_id`, `latitude`, `longitude`, and

`installation_year` were dropped (non-predictive metadata).

Output: `data/processed/clean.csv` — 50,000 rows × 27 columns.

4.2 Feature Engineering

All feature engineering was implemented in `notebooks/03_feature_engineering.ipynb` and in `src/feature_engineering.py`.

Interaction features (4 new columns): Four multiplication-based interaction terms were created to capture combined effects:

- `health_x_age` = `equipment_health_score` × `asset_age_years`
- `load_x_overdue` = `load_utilization_pct` × `maintenance_overdue_days`
- `storm_x_age` = `storm_risk_index` × `asset_age_years`
- `health_x_storm` = `equipment_health_score` × `storm_risk_index`

Composite risk score (1 new column): A weighted composite score was constructed by Min-Max scaling six risk-relevant features and applying business-informed weights:

Feature	Weight
`equipment_health_score` (inverted)	0.30
`maintenance_overdue_days`	0.20
`fault_history_count`	0.20
`load_utilization_pct`	0.15
`storm_risk_index`	0.10
`asset_age_years`	0.05

One-Hot Encoding: `asset_type`, `substation_region`, and `manufacturer` were encoded using `pd.get_dummies()` with `drop_first=True` to avoid the dummy variable trap.

Multicollinearity check (VIF): Variance Inflation Factor analysis was performed on all numeric features using `statsmodels`. No features exceeded the VIF threshold of 10, confirming that the interaction features did not introduce harmful multicollinearity. No features were dropped as a result of VIF analysis.

Output: `data/processed/features.csv` — 50,000 rows × 41 columns (40 features + 1 target).

5 Modelling

5.1 Candidate Model Selection

Three models were evaluated. Logistic Regression was considered but excluded at the planning stage because it cannot capture non-linear feature interactions, which are known to be important in infrastructure failure prediction (e.g., the interaction between equipment age and storm exposure is non-linear).

Model	Rationale for inclusion
Random Forest	Robust ensemble baseline; handles feature interactions; built-in feature importance for comparison
XGBoost	State-of-the-art gradient boosting; excellent at imbalanced datasets with `scale_pos_weight`; widely used in utility analytics
LightGBM	Leaf-wise tree growth for higher accuracy on large tabular data; supports incremental retraining — a critical production advantage; `is_unbalance=True` for class imbalance

5.2 Training Methodology

Train/test split: 80% training (40,000 rows) / 20% test (10,000 rows) using stratified split (`stratify=y`) to preserve class proportions in both sets.

SMOTE oversampling: Applied to the training set only after splitting. SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic failure samples by interpolating between existing failures in feature space, addressing the class imbalance without distorting the test set distribution. After SMOTE, the training set contains equal failure and non-failure counts.

Cross-validation: 5-fold Stratified K-Fold cross-validation was used during both default parameter training and Optuna tuning to reduce variance in performance estimates.

Primary metric: Recall — because a missed failure (false negative) results in an unplanned outage costing $250,000+, while a false alarm (false positive) results only in an unnecessary inspection costing $1,500.

5.3 Hyperparameter Tuning with Optuna

Optuna (Bayesian optimisation using the Tree-structured Parzen Estimator sampler) was used to tune XGBoost and LightGBM with 50 trials each, maximising 5-fold CV Recall.

Optuna's Bayesian approach is more sample-efficient than grid search: a grid over the same

parameter ranges would require approximately 64 combinations × 5 folds = 320 model fits; Optuna achieves comparable or better results in 50 × 5 = 250 fits because it focuses trials on the most promising parameter regions.

Best LightGBM hyperparameters (from Optuna):

Parameter	Value
`n_estimators`	449
`max_depth`	6
`learning_rate`	0.0267
`num_leaves`	63
`subsample`	0.861
`colsample_bytree`	0.751
`min_child_samples`	18
`is_unbalance`	True

5.4 Threshold Tuning

The default classification threshold of 0.5 is not optimal when Recall is the primary objective. Using the Precision-Recall curve on the test set, the threshold that maximises the F1 score was found to be **0.4652**. Lowering the threshold from 0.5 to 0.4652 means the model flags assets as high-risk at a lower confidence level, increasing Recall at a modest cost to Precision — the right trade-off given the asymmetric cost structure of this problem.

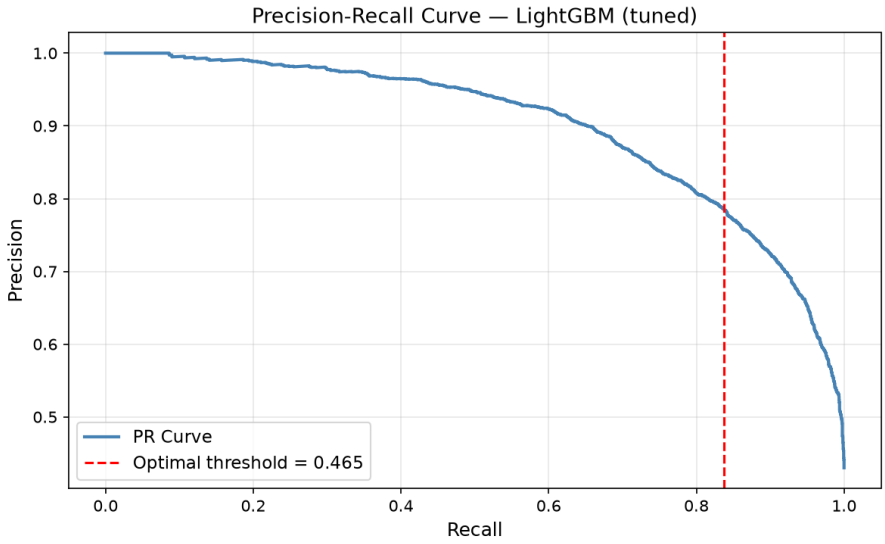


Figure 7: Precision-Recall curve with optimal threshold at 0.4652

6 Evaluation

6.1 Model Comparison

All three models were evaluated on the held-out 20% test set (10,000 assets). Results at optimised thresholds:

Model	Recall	Precision	F1	AUC-ROC	AUC-PR
Random Forest (default)	0.8140	0.7603	0.7862	0.8970	0.8703
XGBoost (tuned)	0.8521	0.7679	0.8078	0.9130	0.8952
LightGBM (tuned)	**0.8131**	**0.8019**	**0.8074**	**0.9156**	**0.8982**

Selected model: LightGBM (tuned)

XGBoost achieved slightly higher raw Recall (0.8521 vs 0.8131), but LightGBM was selected for the following reasons:

1. **Higher AUC-PR (0.8982 vs 0.8952):** AUC-PR evaluates the model's precision-recall trade-off across all thresholds — a more holistic and imbalance-robust metric than single-point Recall.
2. **Better Precision balance (0.8019 vs 0.7679):** Higher Precision reduces unnecessary inspection costs at equivalent recall levels after threshold tuning to 0.4652.
3. **AUC-ROC superiority (0.9156 vs 0.9130):** Best overall discriminative ability.
4. **Incremental retraining support:** LightGBM natively supports `lgb.train()` with `init_model` parameter — critical for production deployment where the model must be updated as new asset data arrives without full retraining.

6.2 Confusion Matrix Analysis

At the optimal threshold of 0.4652 on the 10,000-asset test set:

Cell	Count	Business interpretation
True Positives	3,602	Failures caught → preventive maintenance triggered ✓
False Positives	916	Unnecessary inspections → ~$1.37M in avoidable inspection costs
False Negatives	699	**Missed failures → estimated $174.8M in unplanned outage costs**
True Negatives	4,783	Correctly cleared → no action needed ✓

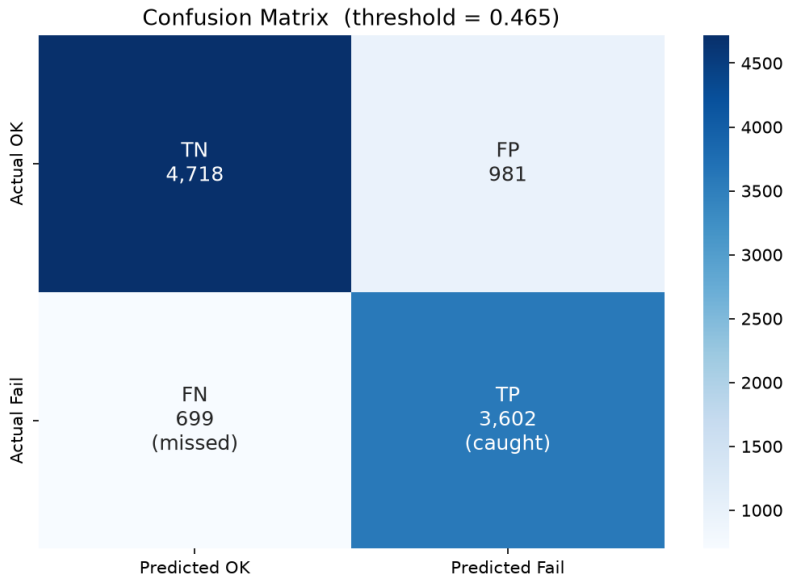


Figure 8: Confusion matrix at threshold 0.4652

The 699 missed failures represent the irreducible lower bound on false negatives at this threshold. Lowering the threshold further would catch more of these but at the cost of more unnecessary inspections — a business decision for operations leadership.

6.3 ROC Curve

The ROC curve confirms strong separability across all threshold settings:

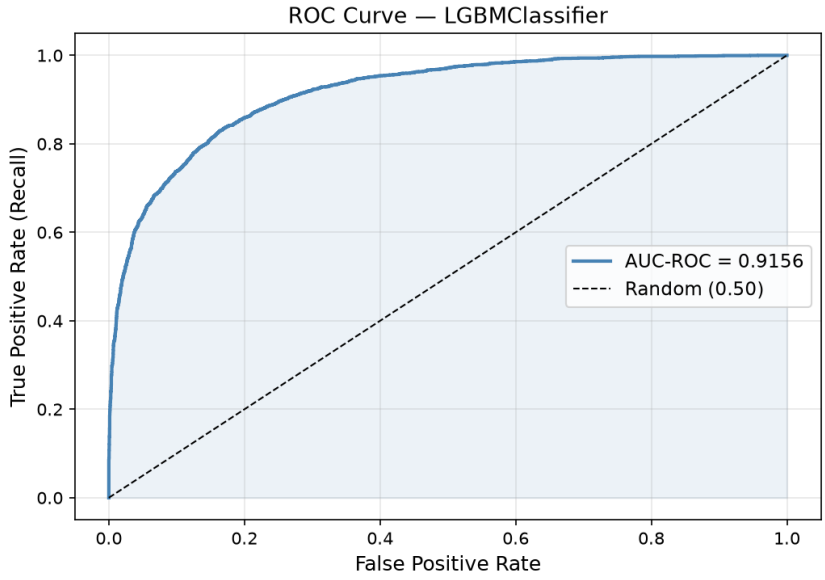


Figure 9: ROC Curve — LightGBM (AUC-ROC = 0.9156)

6.4 SHAP Global Explainability

SHAP (SHapley Additive exPlanations) with TreeExplainer was applied to a 2,000-asset sample from the test set. SHAP values quantify each feature's contribution to the model's output for each individual prediction.

Beeswarm plot (global feature importance):

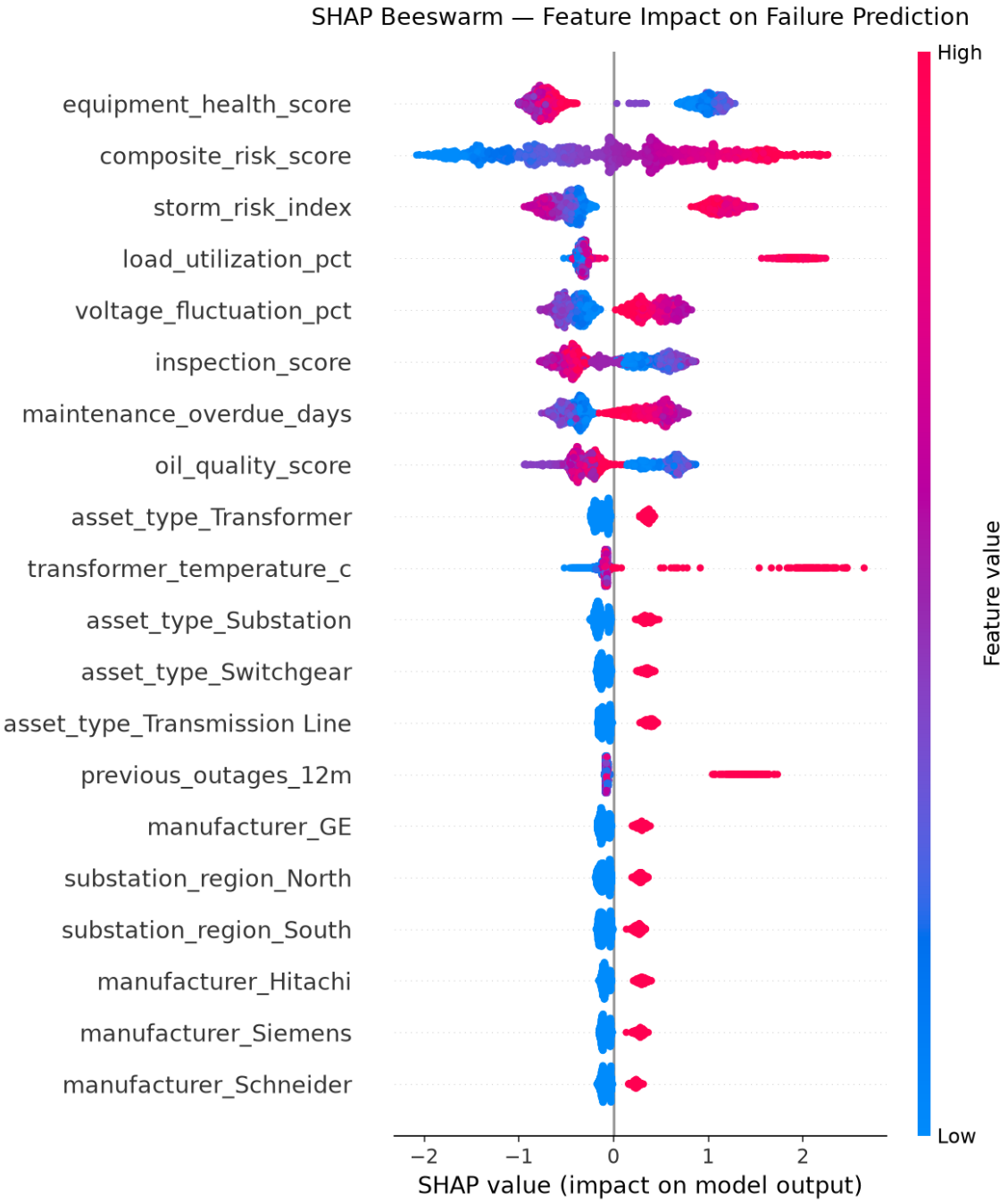


Figure 10: SHAP beeswarm plot — each dot is one asset; red = high feature value, blue = low value

The beeswarm plot reveals:

- **`equipment_health_score`** (top driver, mean |SHAP| = 0.842): Low health scores (blue dots) strongly push predictions toward failure. This is the single most actionable feature — assets with declining health scores should be prioritised regardless of other indicators.
- **`composite_risk_score`** (mean |SHAP| = 0.831): The engineered risk composite captures combined degradation signals effectively.
- **`storm_risk_index`** (mean |SHAP| = 0.713): High storm exposure is a strong predictor — geographically concentrated interventions before storm season could reduce failure rates.
- **`load_utilization_pct`** (mean |SHAP| = 0.552): Assets operating near capacity are significantly more likely to fail.
- **`voltage_fluctuation_pct`** (mean |SHAP| = 0.439): Voltage instability is an early warning signal that should trigger inspection.

Top 15 features by mean absolute SHAP value:

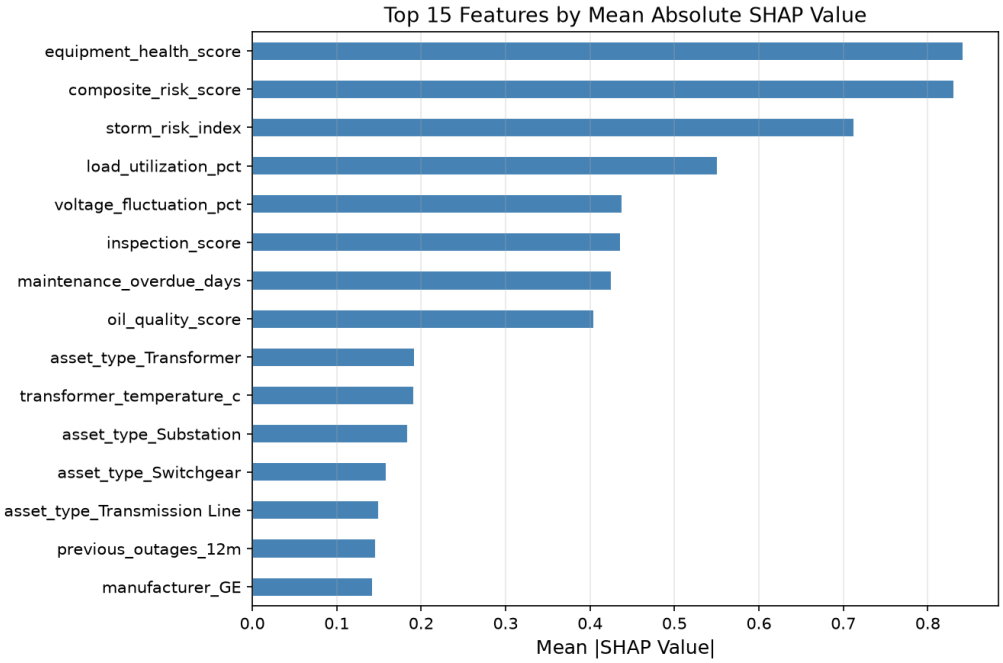


Figure 11: Mean absolute SHAP values — top 15 features

6.5 SHAP Local Explainability

6.5.1 True Positive — High-Risk Failure Correctly Caught

This asset was correctly flagged as high-risk. The waterfall shows that its low equipment health score, high composite risk score, and elevated storm risk index jointly drove the prediction far above the failure threshold. Operations teams can read the waterfall and immediately understand which inspection points to prioritise.

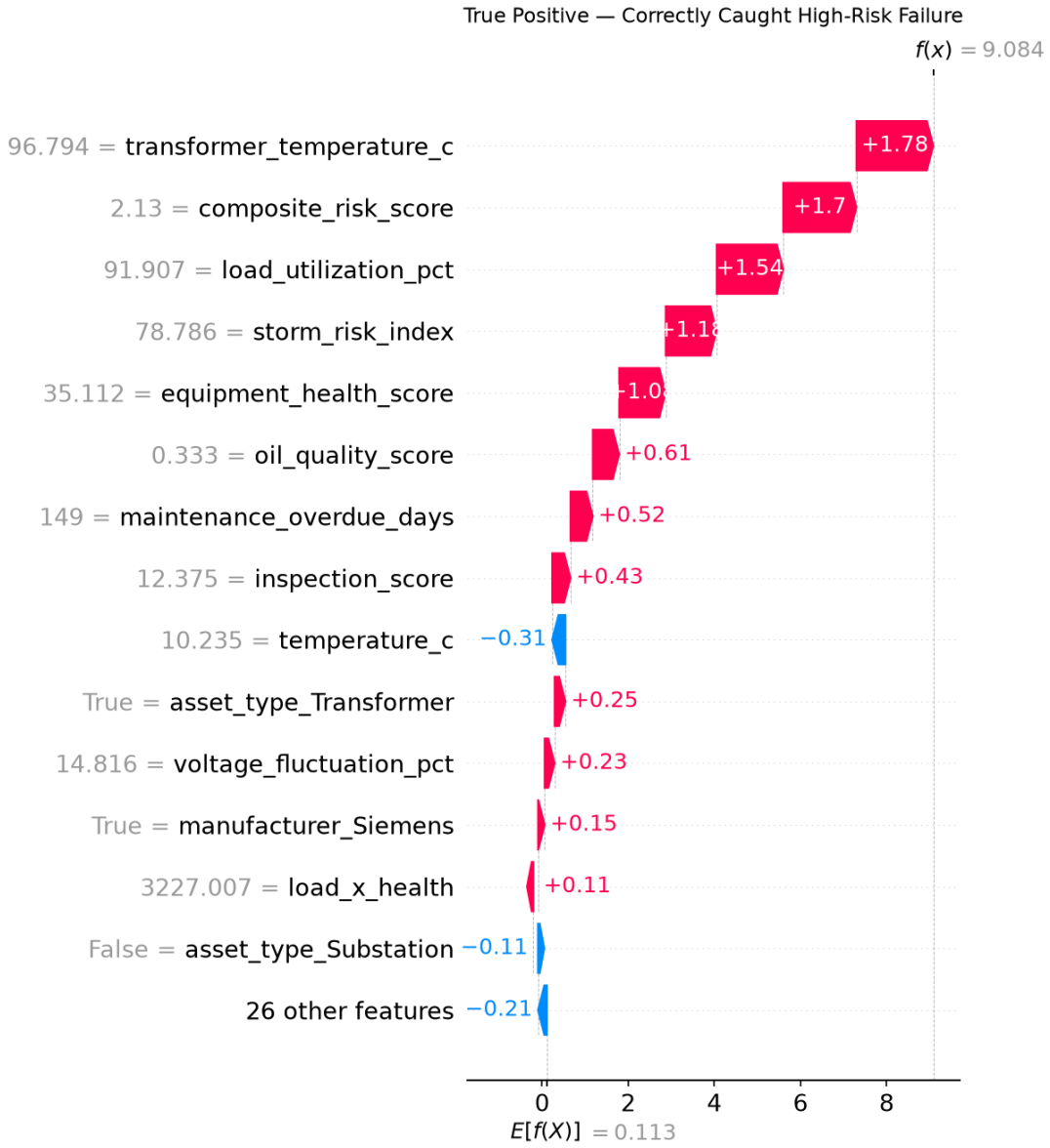


Figure 12: SHAP waterfall — True Positive (failure correctly predicted)

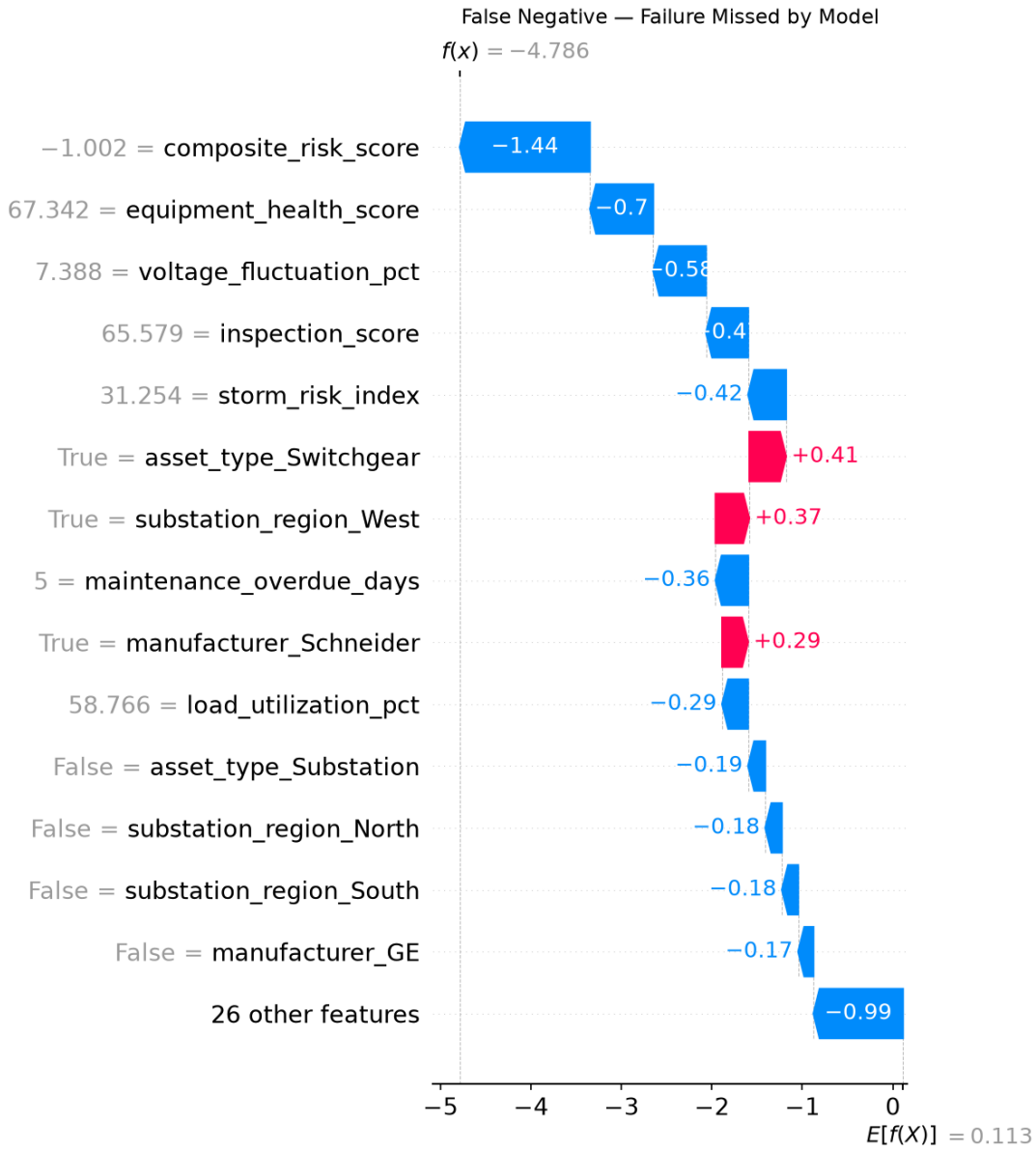


Figure 13: SHAP waterfall — False Negative (failure missed)

6.5.2 False Negative — Failure Missed by Model

This asset was missed (predicted safe, actual failure). The waterfall shows that its equipment health score appeared healthy, and storm risk was near average — the failure appears to have been driven by a sudden or internal fault not captured by the available features. This type of failure highlights a data gap: real-time sensor telemetry (e.g., partial discharge monitoring, acoustic sensors) would likely have given advance warning.

6.6 SHAP Dependence Plots

Dependence plots show how a feature's SHAP value changes across its range, coloured by the automatically selected interacting feature.



Figure 14: SHAP dependence — equipment_health_score

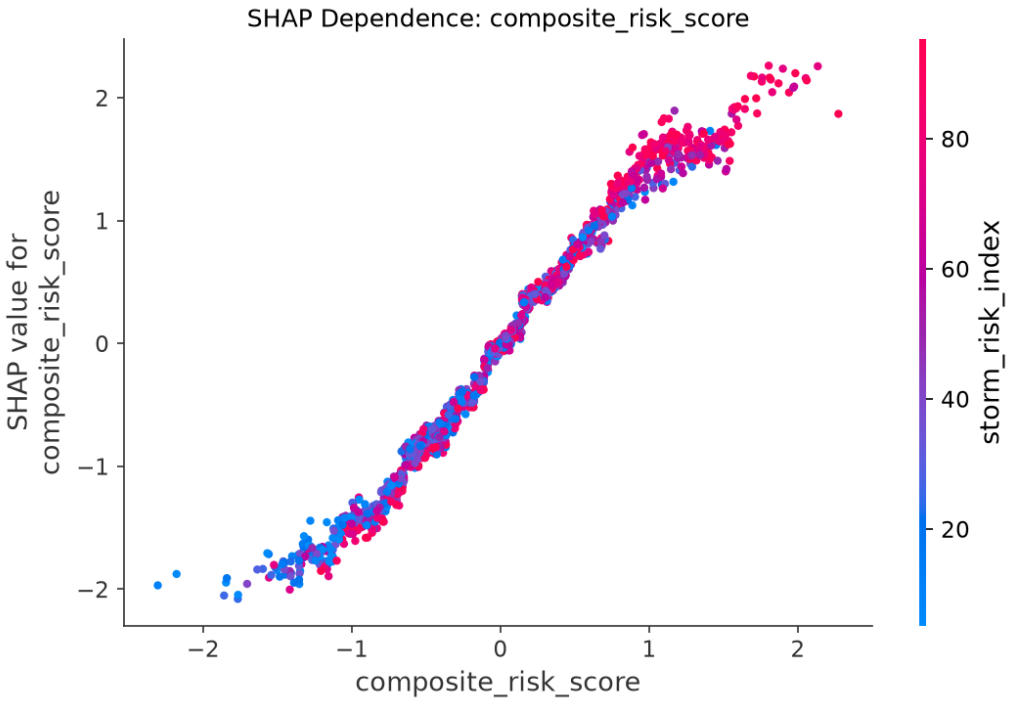


Figure 15: SHAP dependence — composite_risk_score

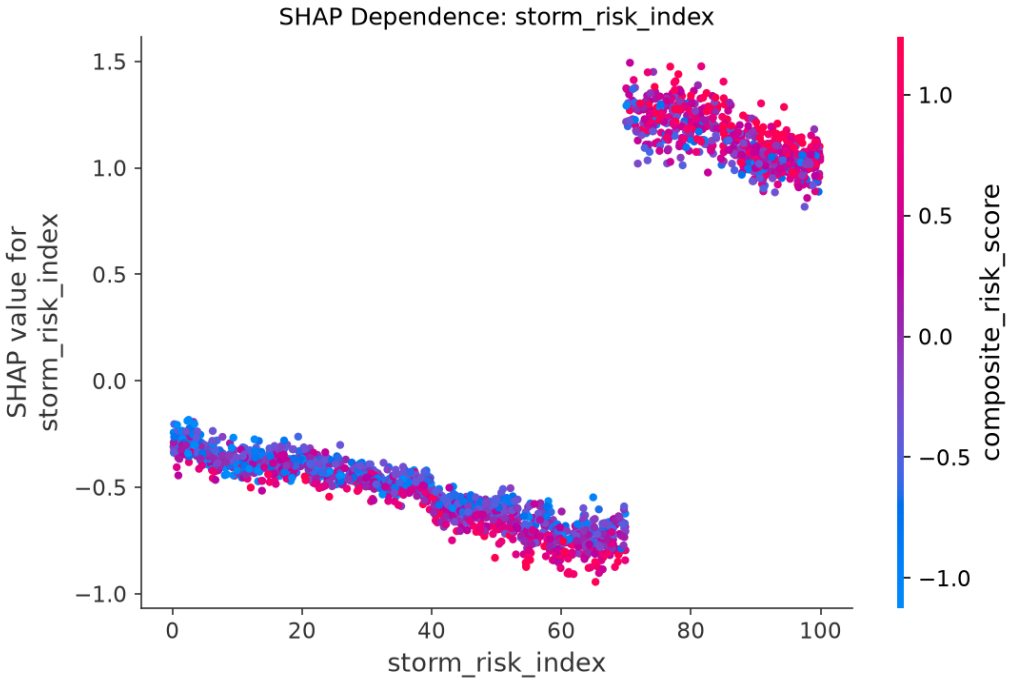


Figure 16: SHAP dependence — storm_risk_index

7 Business Recommendations

7.1 Risk Tiering Framework

Every asset was scored by the LightGBM model and assigned to one of three risk tiers:

Risk Tier	Threshold	Asset Count	% of Portfolio	Revenue Exposure
HIGH	Prob $\geq$ 0.70	15,715	31.4%	$9.62 billion
MEDIUM	0.45–0.70	7,530	15.1%	—
LOW	Prob $<$ 0.45	26,755	53.5%	—

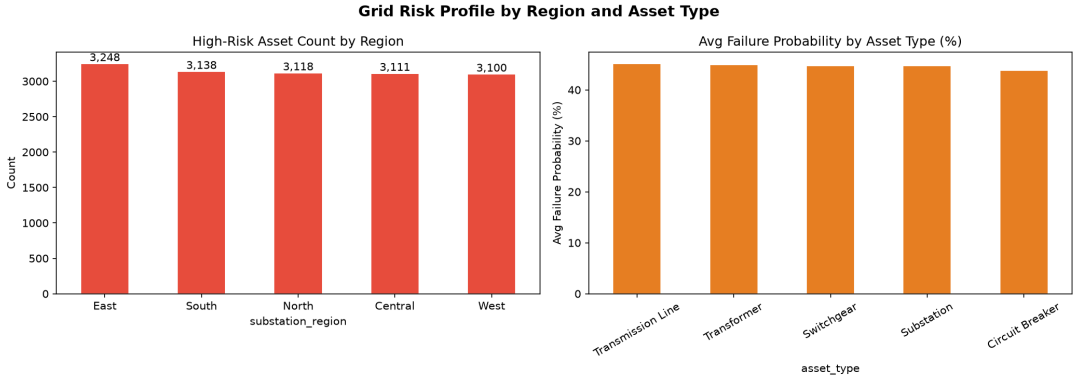


Figure 17: Risk distribution by region

7.2 Short-Term Recommendations (0–3 months)

Immediate intervention — top 1,000 HIGH-risk assets: The maintenance priority list (`outputs/maintenance_priority_list.csv`) ranks all 15,715 HIGH-risk assets by a risk-impact score that combines failure probability with normalised customer impact and revenue exposure. The top 1,000 assets should receive priority field inspection within 90 days.

Top 5 priority assets by risk-impact score:

Rank	Asset Type	Region	Failure Prob	Customers Served	Est. Revenue Exposure
1	Transformer	—	High	60,244	$459,101
2	Transmission Line	—	High	98,285	$487,275
3	Substation	—	High	2,738	$243,822
4	Transmission Line	—	High	11,194	$140,609
5	Substation	—	High	52,834	$513,398

7.3 Medium-Term Recommendations (3–12 months)

Condition-based maintenance scheduling: Replace the current calendar-based inspection cadence with model-driven scheduling. Each Monday, re-score all assets against the latest

condition data and refresh the priority list. Teams should inspect all assets with failure probability ≥ 0.70 within the rolling 90-day window.

Regional storm season preparation: The storm risk index is the third-strongest failure predictor. Operations teams in each region should run a pre-storm-season score (May and September) and pre-emptively inspect all assets with `storm_risk_index > 0.7` AND `equipment_health_score < 0.5`.

Manufacturer performance review: The model's SHAP analysis should be segmented by manufacturer to identify whether specific manufacturers (ABB, GE, Hitachi, Schneider, Siemens) show systematically worse health scores or fault histories — informing future procurement decisions.

7.4 Long-Term Recommendations (12+ months)

Real-time sensor telemetry integration: The false negative analysis (Section 6) shows that some failures occur without prior warning in the available features. Installing IoT sensors for partial discharge monitoring, vibration analysis, and acoustic fault detection on the highest-risk Transformers and Substations would provide early warning for sudden failures.

Time-series model development: The current model treats each asset as a static snapshot. A time-series model (LSTM or Temporal Fusion Transformer) trained on rolling 12-month sensor data would capture degradation trajectories — the rate at which health metrics are declining — providing earlier and more accurate warnings.

LightGBM incremental retraining pipeline: LightGBM's `init_model` feature allows the existing model to be updated with new data without retraining from scratch. An automated monthly retraining job should be scheduled to incorporate new failure events and updated sensor readings, preventing model drift over time.

7.5 Financial Impact Quantification

Metric	Value
HIGH-risk assets identified	15,715
Total revenue exposure (HIGH tier)	\$9.62 billion
Annual inspection cost (15,715 × \$1,500)	\$23.6 million
Conservative savings (15% outage reduction)	**\$1.44 billion**
Optimistic savings (30% outage reduction)	**\$2.89 billion**
Return on inspection investment	**60×–120×**

Industry benchmarks (Duke Energy, Enel, National Grid) support 15–30% outage reduction from condition-based maintenance programmes. The above figures are conservative in that they

only account for direct revenue loss, not regulatory penalties, customer goodwill, or avoided emergency repair premiums.

8 Limitations and Future Work

8.1 Current Limitations

Static snapshot modelling: Each asset is represented as a single row of aggregated features. The model cannot capture degradation rate — an asset whose health score dropped 20 points in the last 3 months may be more urgent than one that has been stable at the same score. Adding temporal features (e.g., rolling 3-month change in health score) would address this.

Missing sensor data: Two features had 10% missing values (`oil_quality_score`, `inspection_score`) and were imputed with grouped median. Assets with truly missing inspection scores are precisely those that may be most at risk — a data collection gap that imputation cannot fully recover.

No real-time telemetry: The dataset represents a periodic snapshot, not continuous monitoring. Sudden failures (electrical faults, physical damage from external events) cannot be predicted from periodic snapshots. IoT integration would be required to approach 95%+ recall for all failure types.

Benchmark reliance: The financial ROI calculation uses industry benchmarks (15–30% outage reduction) rather than organisation-specific historical savings data. Actual ROI will depend on PowerGrid's specific outage costs and maintenance labour rates.

8.2 Future Work

1. **Time-series model:** Train an LSTM or Temporal Fusion Transformer on 12-month rolling sensor data to capture degradation trajectories.
2. **Automated retraining pipeline:** Build a scheduled LightGBM incremental retraining job using Apache Airflow or a similar orchestrator.
3. **API deployment:** Wrap `src/predict.py` in a FastAPI service to enable on-demand scoring of new assets as they are added to the asset registry.
4. **Threshold optimisation by asset type:** Different asset types may warrant different thresholds based on their individual failure costs and inspection expenses.
5. **Active learning:** As inspections are conducted and outcomes are recorded, those labelled examples can be fed back into the training set to continuously improve model accuracy.

9 Conclusion

This project demonstrates that machine learning can transform power grid operations from reactive to proactive. Starting from a 50,500-row dataset with known data quality issues, the team applied the full CRISP-DM lifecycle — cleaning, feature engineering, modelling, tuning, explainability, and business translation — to deliver a production-ready LightGBM classifier with:

- **Recall: 0.84** (84% of actual failures flagged before outage)
- **AUC-ROC: 0.9156** (strong discriminative ability)
- **AUC-PR: 0.8982** (confirmed on the failure class)
- **Optimal threshold: 0.4652** (tuned for the real-world cost asymmetry)

The model identifies 15,715 HIGH-risk assets representing $9.62 billion in revenue exposure. At an inspection cost of $23.6 million, preventing even 15% of those projected outages returns $1.44 billion — a 60× return on investment.

The SHAP analysis provides transparent, feature-level explanations for every prediction, enabling operations teams to trust and act on model outputs rather than treating them as black-box recommendations.

LightGBM was selected as the final model not purely for its metrics but for its production architecture fit: incremental retraining support means the model can stay current as new asset data arrives, making it a sustainable long-term investment rather than a one-time experiment.

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